

EVENT RAKSHAK

UML & Architecture Diagrams

Contains: Use Case, Class, Sequence, Activity (x2), Deployment Diagrams

UML and Architecture Diagrams

This document contains all six UML and architecture diagrams for the Event Rakshak platform. Each diagram is presented on a dedicated page with a title bar, a contextual description, and an italic figure caption. These diagrams are extracted from SRS Appendix E.

Figure 1 — Use Case Diagram

Figure 2 — Class Diagram (Domain Model)

Figure 3 — System Sequence Diagram

Figure 4 — Activity Diagram — Risk Assessment & Subscription

Figure 5 — Activity Diagram — Claims Filing & Adjudication

Figure 6 — Deployment Diagram

1. Use Case Diagram

Captures the full functional scope across Customer, Underwriter, Claims Officer, and System Admin. Use cases are grouped within the Event Rakshak System boundary with <<include>> relationships for shared sub-functions: Risk Calculation, Weather Data Fetch, and Upload Evidence.

Figure 1 — Use Case Diagram

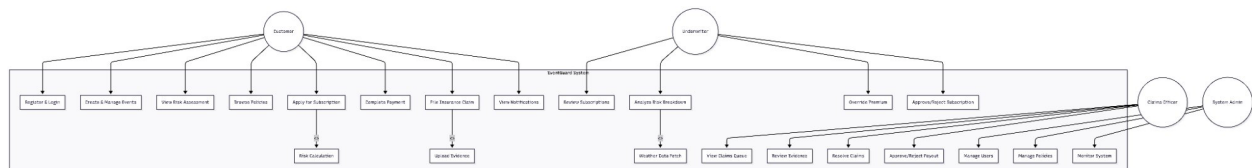


Figure 1 — Use Case Diagram

2. Class Diagram (Domain Model)

Shows all primary Java entity classes (User, Event, Policy, PolicySubscription, Risk, Claim, Notification) with attributes, data types, and inter-class associations. Enumeration types (EventDomain, VenueType, SubscriptionStatus, ClaimStatus) are shown with multiplicities.

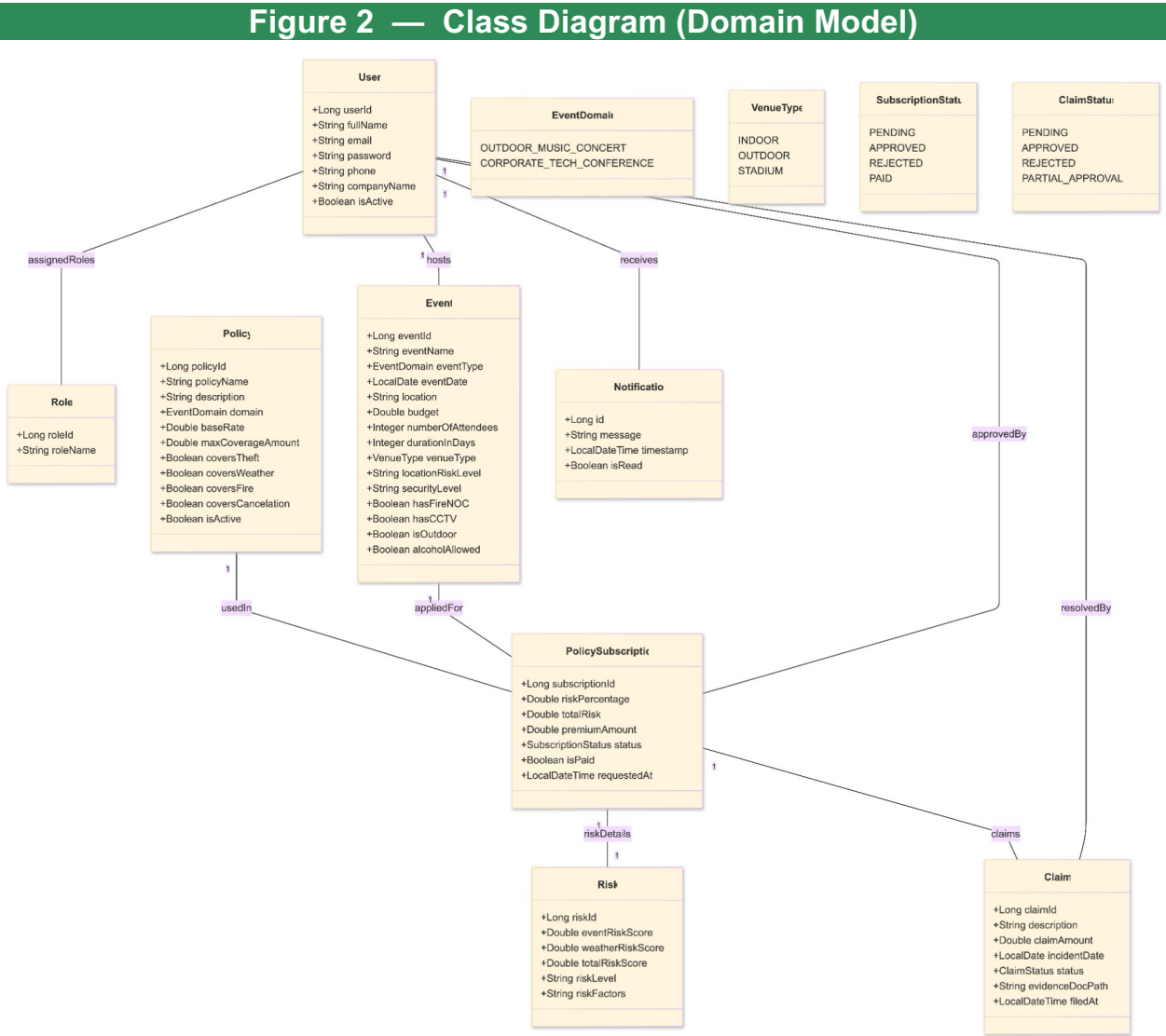


Figure 2 — Class Diagram (Domain Model)

3. System Sequence Diagram

Full end-to-end message flow across: Customer, Web Portal (Angular 21), API Gateway/Auth, Event & Risk Service, WeatherAPI, Insurance Underwriter, Claims Officer, and H2 Database. Covers login, event creation, risk calculation, subscription, payment, claims, and notifications.

Figure 3 — System Sequence Diagram

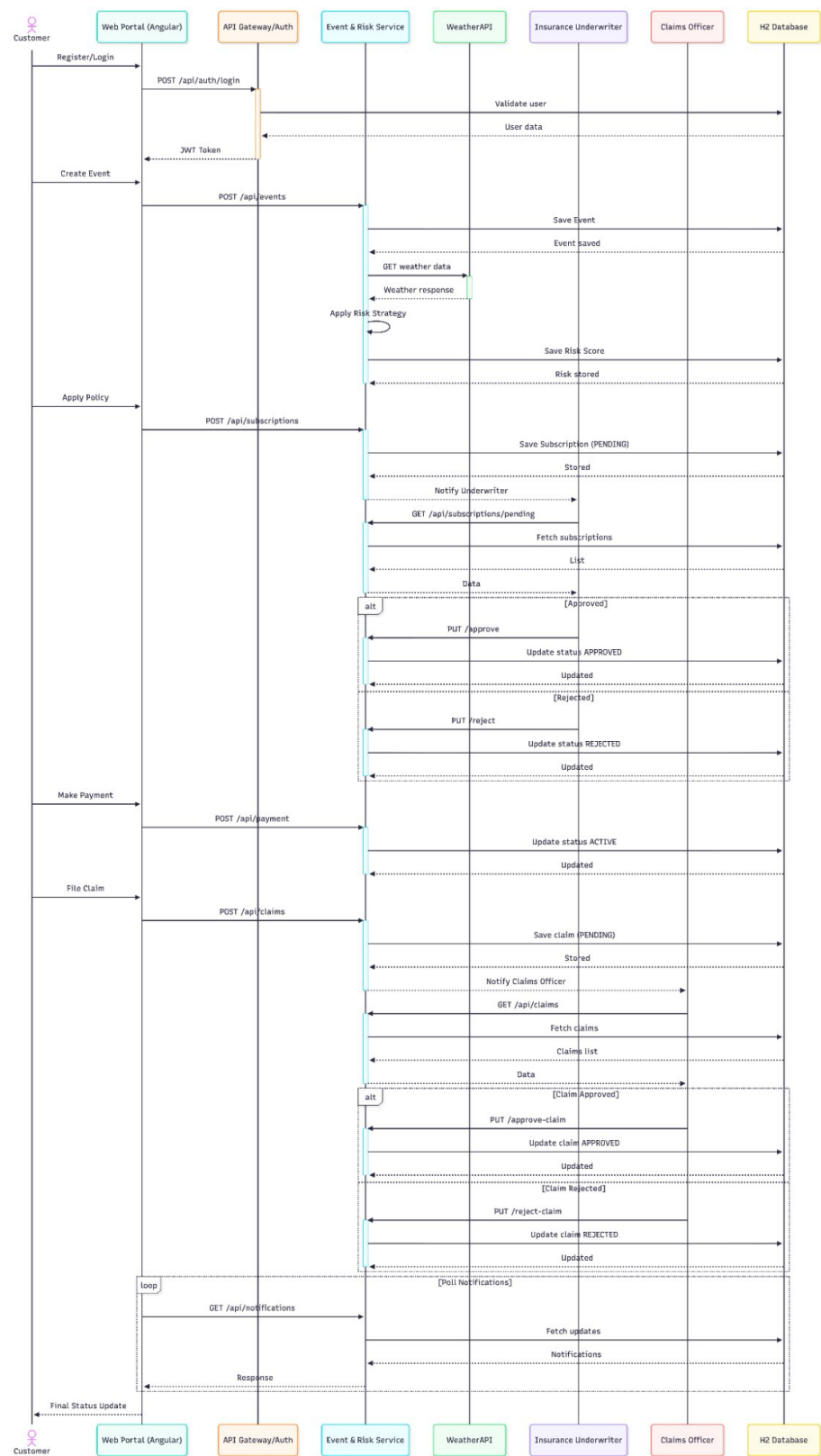


Figure 3 — System Sequence Diagram

4. Activity Diagram — Risk Assessment & Subscription

Models the flow from event detail entry through conditional WeatherRiskService invocation (decision node s11 for outdoor events), domain penalty/mitigation application via the Strategy Pattern, underwriter approve/reject decision (s12), payment, and Subscription ACTIVE status.

Figure 4 — Activity Diagram — Risk Assessment & Subscription

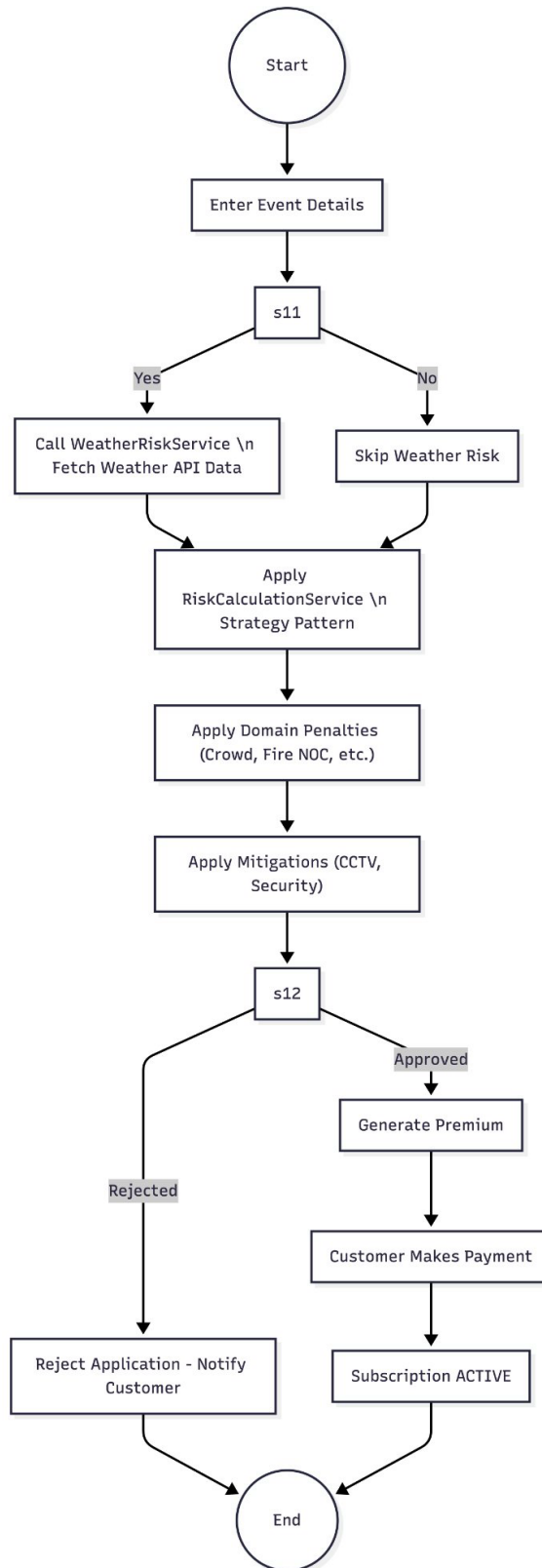


Figure 4 — Activity Diagram — Risk Assessment & Subscription

5. Activity Diagram — Claims Filing & Adjudication

Models the post-incident claims workflow. Decision node s9 validates the claim; s10 branches on the Claims Officer's determination. Approved path triggers payout entry and RESOLVED status; rejected path captures a mandatory rejection reason.

Figure 5 — Activity Diagram — Claims Filing & Adjudication

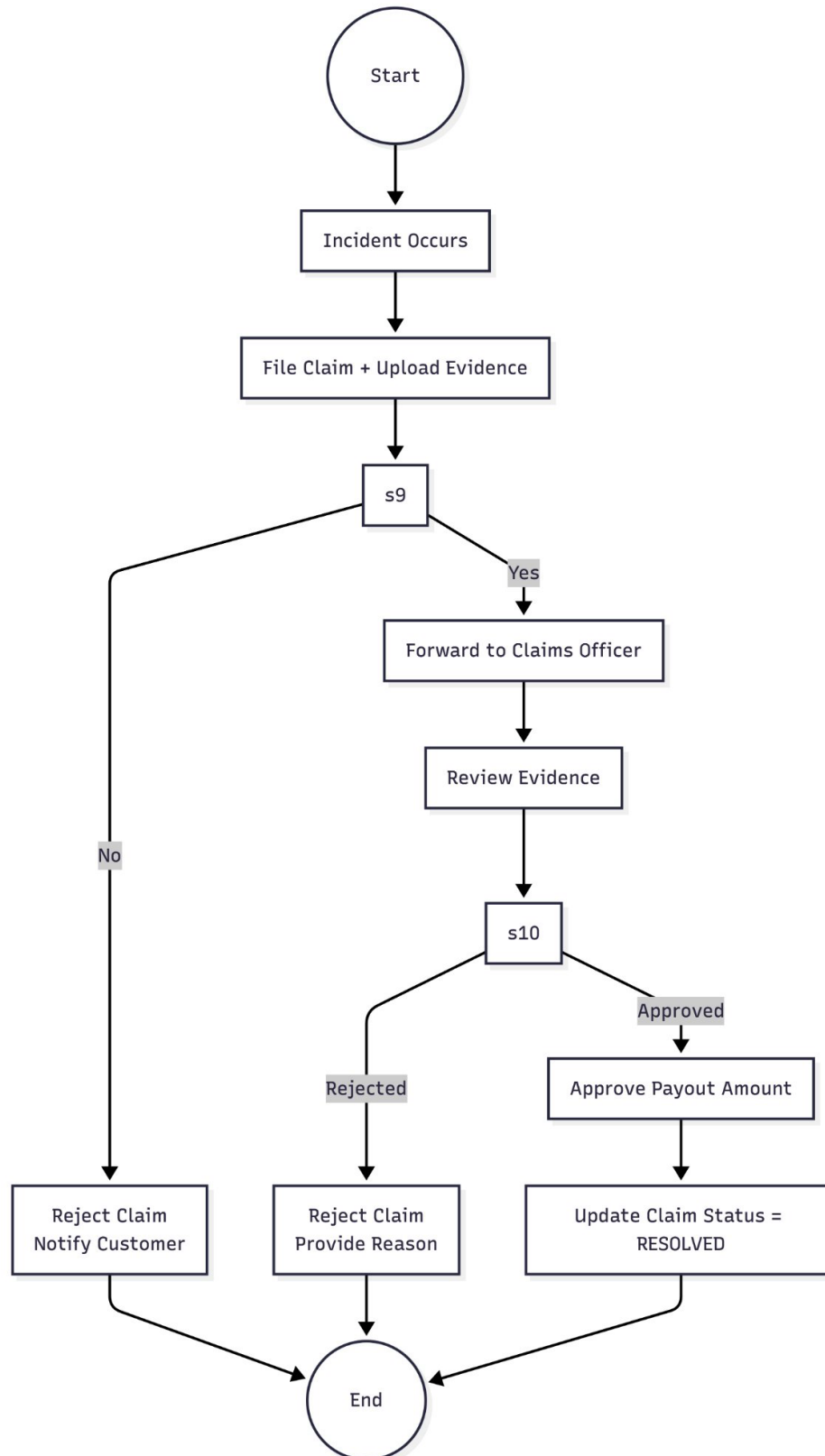


Figure 5 — Activity Diagram — Claims Filing & Adjudication

6. Deployment Diagram

Illustrates Developer Workstation → CI/CD (Jenkins/GitHub Actions with JUnit 5 and Karma/Jasmine parallel test suites) → Production Web Server (Nginx → TLS → Spring Boot API Gateway → H2 persistent file-based storage). WeatherAPI.com shown as third-party integration.

Figure 6 — Deployment Diagram

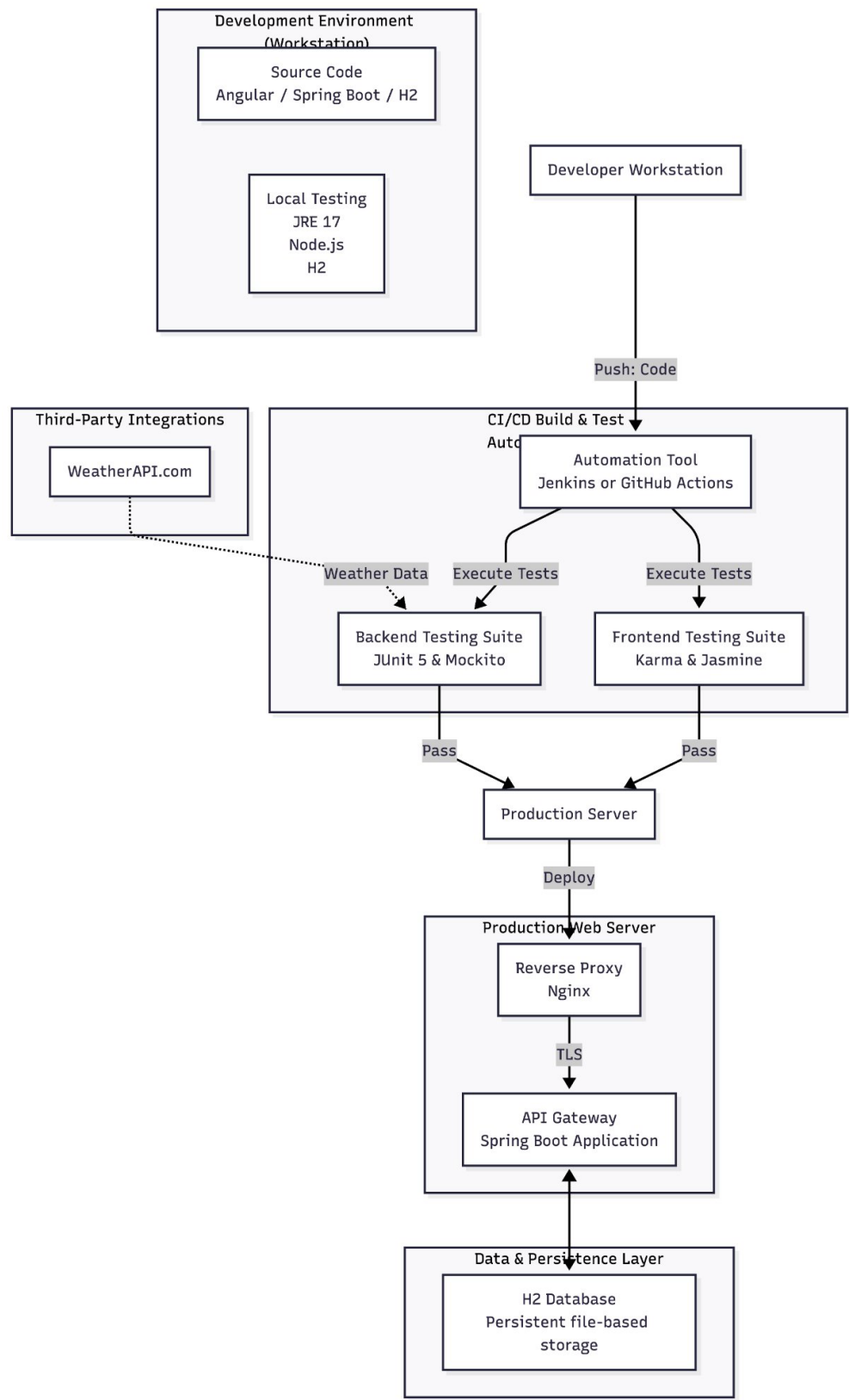


Figure 6 — Deployment Diagram